TUNISIA REAL ESTATE PRICE PREDICTION

A Data Mining Analysis of the Tunisian Property Market
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PROBLEM STATEMENT

The Tunisian real estate market suffers from significant information asymmetry. Buyers and sellers often have very different knowledge of what a fair price looks like for a given property.

Task 1 - Regression

Objective: Predict the exact listing price (TND)

Research question: Which property attributes best explain price variation?

Target variable: price (continuous, log-transformed)

Task 2 - Classification

Objective: Assign properties to one of 4 price tiers

Research question: Can we segment the market into Low /Medium / High / Luxury?

Target variable: price_tier (4-class categorical)

DATASET DESCRIPTION

Overview

Context: Tunisian real estate listings scraped from two major portals

Sources: menzili.tn & tunisie-annonce.com

Unit of observation: One property listing published by a seller or agency

Scope: All Tunisian governorates - apartments, villas, duplexes, studios, penthouses..

1400
Observations

15
Variables

Key Variables

price: target variable (TND)
surface_habitable: living area (m2)
c**hambres / salles_de_bain**: rooms
residence_type: apartment, villa, etc.
neighborhood: area in Tunisia
Amenity flag: pool, terrace, elevator ...
cuisine_type: kitchen quality

DATA PREPARATION

1 Missing Value Imputation

Numeric columns filled with column medians, cuisine_type filled with mode ('non equipee')

2 Outlier Removal

7 removed using domain-knowledge bounds: price 60k-2.5M TND,surface 20-500 m2, ≤8 bedrooms, ≤15 total rooms.

3 Feature Engineering

3 new features: space_per_room (comfort index), luxury_score (0-6 amenity sum), neigh_price_index (median price)

4 Categorical Encoding

Ordinal encoding (1-3) for cuisine_type, Neighbourhood encoded as neigh_price_index to capture location prestige.

5 Train / Val / Test Split

70 / 15 / 15 stratified split on price tier - ensures all 4 tiers appear in every partition

6 Feature Scaling

StandardScaler fitted on the training set only. Validation and test sets transformed using training-set parameters

1

METHODS & ANALYSIS

Regression - Predicting Price

KNN (k=10)

Non-parametric: no assumption on price shape. Predicts by averaging 10 nearest neighbours in scaled 20-dim space.

Decision Tree (depth=12)

Scale-invariant; handles mixed feature types natively (booleans, ordinals, continuous).

Random Forest (300 trees)

Ensemble of 300 bootstrapped trees: decorrelates errors & drastically
reduces variance vs. a single tree.

1

RESULTS & INTERPRETATION

Regression - Predicting Price

Regression Performance

	Model	R2	RMSE
0	Random Forest	0.852	134005
1	Decision Tree	0.696	192046
2	KNN (k=10)	0.641	208440

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Best model: Random Forest
Test R²   = 0.747  |  Test RMSE = 154,936 TND
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2

METHODS & ANALYSIS

Classification - Assigning Price Tier

Naive Bayes

Fast baseline under the independence assumption. **Included to demonstrate what happens when model assumptions are violated.**

Random Forest (300 trees)

Bootstrap sampling handles class imbalance naturally. Split-based partitioning exploits feature interactions without normality assumptions.

2

RESULTS & INTERPRETATION

Classification - Assigning Price Tier

Classification Performance

	Model	Accuracy	Precision	Recall	F1
0	Random Forest	0.651	0.643	0.651	0.637
1	Naive Bayes	0.301	0.235	0.301	0.236

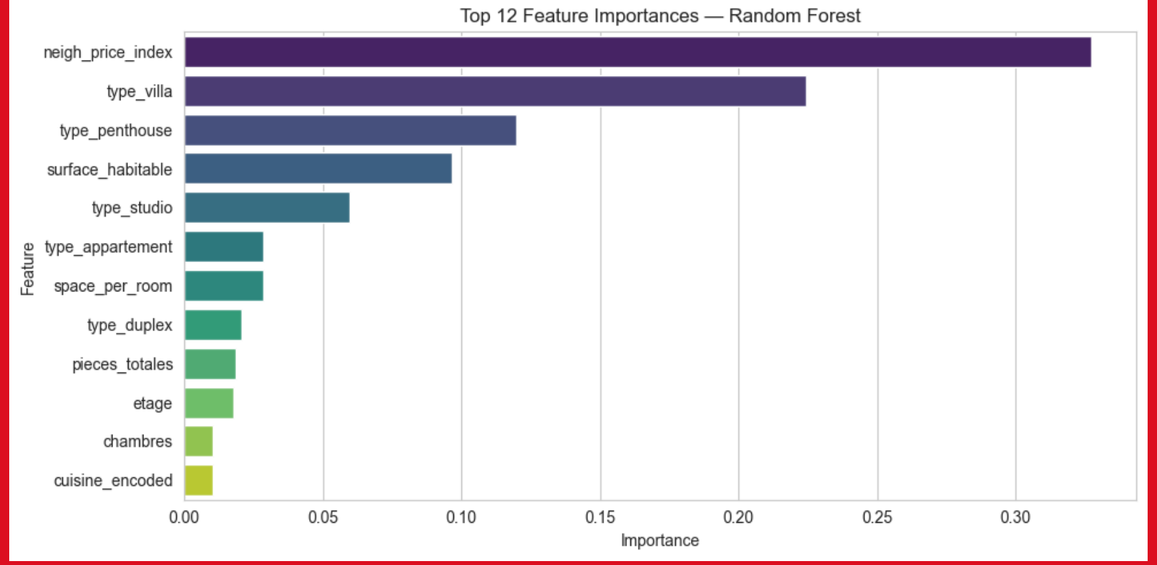
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Best model: Random Forest
Accuracy=0.746  Precision=0.75  Recall=0.746  F1=0.745
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2

KEY FINDINGS

- Random Forest wins both tasks - ensemble variance reduction outperforms KNN and single Decision Tree.
- Location (neigh_price_index) is the #1 price driver at 32.7% importance - more than size, type & amenities combined.
- Property type is the #2 signal - villa + penthouse dummies together account for 34.4% of importance.
- Swimming pool has the largest amenity premium. Properties with a pool have a median price roughly twice that of those without .

Rank	Feature	Importance
1	neigh_price_index	32.7%
2	type_villa	22.4%
3	type_penthouse	12.0%
4	surface_habitable	9.7%
5	type_studio	6.0%



CONCLUSION



This project applied data mining techniques to 1,400 Tunisian real estate listings to build and evaluate price prediction models

Three regression models and two classifiers were benchmarked with the best model evaluated on the test set:

Best regressor: Random Forest , R2 = 0.747, RMSE ≈155,000 TN

Best classifier: Random Forest, Accuracy = 74.6%

LIVE DEMO

Input

A 3-bedroom, 120 m2 apartment in El Manar 2 with a balcony and elevator, on the 2nd floor, with an equipped kitchen.


THANK YOU